Regression Techniques
Lecture Notes

Indian Statistical Institute

Contents

Chapter 1

Pre-requisites

1.1 Introduction to Linear Regression Models.

Data:

- Dependent variable or response (y),
- Independent variables or covariates or regressors $(x_1, x_2, \ldots, x_p)$
- Usually, we have n observations on all variables.

$$\underset{\sim n}{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} : \text{The vector of } n \text{ observations of the response.}$$

$$\begin{aligned} X_n &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}^{n\times p} \quad : \text{Design matrix or matrix of } n \text{ observations of } p \text{ covariates} \\ &= ((x_{ij})) \end{aligned}$$

$\underset{\sim i}{x}$: ith row of X_n provides the ith observation on all variables, $i = 1, \ldots, n$.
$\underset{\sim j}{x}$: jth column of X_n provides a vector of all observations on the jth variables, $j = 1, \ldots, p$.

Examples
1. Air-quality (UCI). Data on 5 metal oxide chemical sensors (CO, Non-metanic hydrocarbon, Benzene, NOx, NO_2) and the temperature, humidity, relative humidity. (See the data here)
Q: Does air-quality affect the temperature, or humidity, or relative humidity?

2. Boston Housing data (Kaggle). See the data description here. Read the data description and identify the underlying regression-related question in the data.

Regression setup. How to relate the temperature with the air pollution indicators in the AQI data?

Usually all the variables are assumed to be random. At least the response variable (Y) is assumed to be random, and the n observations $\underset{\sim n}{y} = (y_1, \ldots, y_n)^\top$ are realizations of n independent samples.

In regression, one assumes that

$$E(Y) = g(x_1, \ldots, x_p), \quad \text{or,} \quad E(y|X_1 = x_1, \ldots, X_p = x_p) = g(x_1, \ldots, x_p),$$

according as the covariates are non-random, or random.

Linear Model

In linear model, we assume

$$E(Y) = g(x_1, \ldots, x_p) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

where $\beta_0, \beta_1, \ldots, \beta_p$ are unknown constants or parameters, to be estimated using the data.

Here linear means, linear in parameters. A function, which is not linear in $\{x_1, \ldots, x_p\}$ can be a linear regression model. For example,

$$E(Y) = \beta_0 + \beta_1 \sin x_1 + \beta_2 e^{x_2}$$

is also a linear model.

Note: In simple linear model literature, $\mathrm{Var}(Y) = \sigma^2$, which is free of $x_1, \ldots, x_p$.

What can be said about joint distribution of $Y_1, \ldots, Y_n$?

Random vectors. Let

$$\underset{\sim}{Z} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$$

be a random vector, i.e., $Z_1, \ldots, Z_n$ are random variables and $\underset{\sim}{Z}$ is a stacked version of all of these random variables.

$$\begin{aligned} f_{\underset{\sim}{Z}}(\underset{\sim}{z}) &= f_{Z_1,\ldots,Z_n}(z_1, \ldots, z_n) \\ &= \begin{cases} P(Z_1 = z_1, \ldots, Z_n = z_n) & \text{for discrete random variables,} \\ \text{joint pdf of } \{Z_1, \ldots, Z_p\} \text{ evaluated at } \{z_1, \ldots, z_p\}. \end{cases} \end{aligned}$$

Expectation and Variance.

$$E(\underset{\sim}{Z}) = \begin{bmatrix} E(Z_1) \\ \vdots \\ E(Z_n) \end{bmatrix} = \underset{\sim Z}{\mu}$$

Let $\underset{\sim}{Z} \in \mathbb{R}^p$ and $\underset{\sim}{W} \in \mathbb{R}^q$ be two random vectors, then

$$\begin{aligned} \mathrm{Cov}(\underset{\sim}{Z}, \underset{\sim}{W}) &= ((\mathrm{Cov}(Z_j, W_k)))^{p \times q}, \qquad j = 1, \ldots, p;\ k = 1, \ldots, q, \\ \mathrm{Var}(\underset{\sim}{Z}) &= E\left[(\underset{\sim}{Z} - \underset{\sim Z}{\mu})(\underset{\sim}{Z} - \underset{\sim Z}{\mu})^\top\right] = ((\mathrm{Cov}(Z_i, Z_j))) = \mathrm{Cov}(\underset{\sim}{Z}, \underset{\sim}{Z}) = \mathrm{Var}(\underset{\sim}{Z}) \end{aligned}$$

Note:

1. $E(\underset{\sim}{Z})$ is a non-random vectors, and $\mathrm{Var}(\underset{\sim}{Z})$ is a non-random $p \times p$ matrix.
2. $\mathrm{Var}(\underset{\sim}{Z})$ is a positive semi-definite matrix.

Some exercises:

Show the following:

1. $\mathrm{Var}(\underset{\sim}{Z} - \underset{\sim}{\beta}) = \mathrm{Var}(\underset{\sim}{Z})$.

2. $\mathrm{Cov}(\underset{\sim}{a}^\top \underset{\sim}{Z}, \underset{\sim}{b}^\top \underset{\sim}{W}) = \underset{\sim}{a}^\top \mathrm{Cov}(\underset{\sim}{Z}, \underset{\sim}{W})\underset{\sim}{b}$.

3. $\mathrm{Cov}(\underset{\sim}{X}_1 + \underset{\sim}{X}_2, \underset{\sim}{Z}_1 + \underset{\sim}{Z}_2) = \mathrm{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_1) + \mathrm{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_2) + \mathrm{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_1) + \mathrm{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_2)$.

4. $\mathrm{Var}(\underset{\sim}{X} \pm \underset{\sim}{Z}) = \mathrm{Var}(\underset{\sim}{X}) + \mathrm{Var}(\underset{\sim}{Z}) \pm \mathrm{Cov}(\underset{\sim}{X}, \underset{\sim}{Z}) \pm \mathrm{Cov}(\underset{\sim}{Z}, \underset{\sim}{X})$.

5. $E(\underset{\sim}{Z}^\top A\underset{\sim}{Z}) = \underset{\sim}{\mu}^\top A\underset{\sim}{Z} + \mathrm{tr}(A\Sigma)$, where $E(\underset{\sim}{Z}) = \underset{\sim}{\mu}$, $\mathrm{Var}(\underset{\sim}{Z}) = \Sigma$.

Back to the linear regression model:

If $Y_i \sim F(\mu_i, \sigma^2)$ some distribution with mean μ_i and variance σ^2, then $\mu_i = \beta_0 + \beta_1 x_{i,1} + \ldots + \beta_p x_{i,p}$. Further, it is *assumed* that $Y_1, \ldots, Y_n$ are independently distributed. Therefore,

$$E(\underset{\sim}{Y}_n) = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = X_n \underset{\sim}{\beta} \quad \text{and} \quad \mathrm{Var}(\underset{\sim}{Y}_n) = \sigma^2 I_n,$$

where $\underset{\sim}{\beta} = (\beta_0, \ldots, \beta_p)^\top$ is the vector of regression parameters, and I_n is the n-dimensional identity matrix.

Alternative representation. One can rephrase the above model in the following way:

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \ldots + \beta_p x_{i,p} + \varepsilon_i,$$

where $E(\varepsilon_i) = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$ for all $i = 1, \ldots, n$.

1.2 Vector Differentiation

1.2.1 Gradient and Hessian

Let $g : \mathbb{R}^k \to \mathbb{R}$.

Example: i. $k = 2$, $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x + y$. ii. For general k, let $\underset{\sim}{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$, $g\left(\underset{\sim}{x}\right) = x_1 + \cdots + x_k \in \mathbb{R}$.

- **Gradient:**

$$\nabla g = \frac{\partial g}{\partial x} = \begin{bmatrix} \partial g / \partial x_1 \\ \vdots \\ \partial g / \partial x_k \end{bmatrix}^{k \times 1}$$

For $g\left(\underset{\sim}{x}\right) = x_1 + \cdots + x_k$,

$$\nabla g = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1}$$

- **Second derivative / Hessian:** The matrix of second derivatives is called the Hessian matrix, and is defined as

$$H\left(\underset{\sim}{x}\right) = \left(\left(\frac{\partial^2 g}{\partial x_i \partial x_j}\right)\right) = \frac{\partial^2 g}{\partial \underset{\sim}{x} \partial \underset{\sim}{x}^\top} = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \cdots & \frac{\partial^2 g}{\partial x_1 \partial x_k} \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_2 \partial x_k} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 g}{\partial x_k \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_k^2} \end{bmatrix}^{k \times k}.$$

- For vector-valued $g : \mathbb{R}^k \to \mathbb{R}^k$, $g\left(\underset{\sim}{x}\right) = \begin{bmatrix} g_1\left(\underset{\sim}{x}\right) \\ \vdots \\ g_k\left(\underset{\sim}{x}\right) \end{bmatrix}$

$$\nabla g = \left[\left(\frac{\partial g_1}{\partial \underset{\sim}{x}}\right) \cdots \left(\frac{\partial g_k}{\partial \underset{\sim}{x}}\right)\right]^{k \times k} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_k}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_k} & \cdots & \frac{\partial g_k}{\partial x_k} \end{bmatrix}$$

Example: $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$, then

$$\nabla g = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$$

1.2.2 Use 1: Jacobian of Transformation

Recall the transformation technique of obtaining the distribution of a one-to-one function of X, when the distribution of X is known. Now, we will extend the transformation technique to mutivariate change of variables problems.

Univariate: $X \sim f_X(x)$ (pdf), $Z = h(X)$, then

$$f_Z(z) = f_X(x)\Big|_{x \to z} |J_{x\to z}| = f_X\big(h^{-1}(z)\big)\left|\frac{\partial x}{\partial z}\right|.$$

Multivariate: $\underset{\sim}{X} \sim f_{\underset{\sim}{X}}$ (joint pdf), $\underset{\sim}{Z} = h\left(\underset{\sim}{X}\right)$ where $h : \mathbb{R}^p \to \mathbb{R}^p$ is one-to-one function. The pdf of $\underset{\sim}{Z}$ at $\underset{\sim}{z}$ is

$$f_{\underset{\sim}{Z}}(\underset{\sim}{z}) = f_{\underset{\sim}{X}}(\underset{\sim}{x})\Big|_{\underset{\sim}{x} \to \underset{\sim}{z}} |J_{x\to z}| = f_{\underset{\sim}{X}}\big(h^{-1}(\underset{\sim}{z})\big)\left|\det\left(\frac{\partial \underset{\sim}{x}}{\partial \underset{\sim}{z}}\right)\right|.$$

Example 1. Let $X_i \overset{iid}{\sim} N(0, \sigma^2)$, $i = 1, 2$. Let

$$Z = \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}, \qquad Z = AX, \qquad A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$f_{\underset{\sim}{X}}(\underset{\sim}{x}) = \frac{1}{2\pi\sigma^2}\exp\left\{-\frac{1}{2\sigma^2}(x_1^2 + x_2^2)\right\}$$

We have

$$\underset{\sim}{X} = \begin{pmatrix} (z_1 + z_2)/2 \\ (z_1 - z_2)/2 \end{pmatrix} = B\underset{\sim}{Z},$$

$$J_{x\to z} = \left|\det\left(\frac{\partial \underset{\sim}{x}}{\partial \underset{\sim}{z}}\right)\right| = \left|\det\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}\right| = \frac{1}{2}.$$

Thus,

$$f_{\underset{\sim}{Z}}(\underset{\sim}{z}) = \frac{1}{2\pi\sigma^2}\exp\left\{-\frac{1}{2\sigma^2}\left(\frac{(z_1+z_2)^2}{4} + \frac{(z_1-z_2)^2}{4}\right)\right\} \times \frac{1}{2}, \qquad z_1 \in \mathbb{R},\ z_2 \in \mathbb{R},\ \sigma^2 > 0.$$

Identify the distribution of $\underset{\sim}{Z}$.

1.2.3 Use 2: Finding maxima/minima of a multivariate function

Let $g : \mathbb{R}^k \to \mathbb{R}$ function, i.e., $g(\underset{\sim}{z}) = g(z_1, \dots, z_k)$. We are interested in finding the maxima/minima of g w.r.t. $\underset{\sim}{z}$.

Let g is maximized (minimized) w.r.t. $\underset{\sim}{z}$ at $\underset{\sim 0}{z}$, where $\underset{\sim 0}{z}$ is on 'interior point' of the domain. The following steps may lead to the identification $\underset{\sim 0}{z}$:

1. Differentiate g w.r.t. $\underset{\sim}{z}$, and equate that to $\underset{\sim}{0}$, $\dfrac{\partial g}{\partial \underset{\sim}{z}} = \underset{\sim}{0}$.
2. Find the points for which this equation is satisfied. Let $\underset{\sim 1}{x}, \dots, \underset{\sim m}{x}$ be the points. These are called stationary points, or critical points.

   [Note:] A critical point can be maxima, minima or a saddle point.

3. Next evaluate the Hessian $\left.\dfrac{\partial^2 g}{\partial \underset{\sim}{z} \partial \underset{\sim}{z}^\top}\right|_{\underset{\sim}{x}_j} = H(\underset{\sim}{x}_j), \ j = 1, \ldots, k.$

If $\underset{\sim}{x}_0$ is a maxima (minima) then $H(\underset{\sim}{x}_0)$ will be negative (positive) definite.

Example 2. $g : \mathbb{R}^k \to \mathbb{R}$,

$$g\left(\underset{\sim}{x}\right) = \sum_{j=1}^{k} \left(x_j^4 - 4x_j^2\right).$$

i)

$$\nabla g = \begin{bmatrix} 4x_1^3 - 8x_1 \\ \vdots \\ 4x_k^3 - 8x_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Solutions to the first order equations: $x_j \in \{0, \sqrt{2}, -\sqrt{2}\}$. So total 3^k critical points:

$$\underset{\sim}{x}_1^* = \underset{\sim}{0}, \quad \underset{\sim}{x}_2^* = \sqrt{2}\,\underset{\sim}{1}, \quad \underset{\sim}{x}_3^* = \begin{pmatrix} \sqrt{2} \\ \vdots \\ -\sqrt{2} \end{pmatrix}, \quad \ldots, \quad x_m^* = -\sqrt{2}\,\mathbf{1}, \quad x_{m+1}^* = \begin{pmatrix} 0 \\ \sqrt{2} \\ \vdots \\ \sqrt{2} \end{pmatrix}, \quad \ldots$$

ii)

$$H\left(\underset{\sim}{x}\right) = \mathrm{Diag}(12x_1^2 - 8, \ldots, 12x_k^2 - 8)$$

Therefore, $H(\mathbf{0}) = -8I$: negative definite, which implies that $\mathbf{0}$ is a local maxima.

For all combinations with $x_j = \pm\sqrt{2} \ \forall j$, $H = 16I$: positive definite, and therefore, all these points are local minimas.

Note: Points with mix of 0 and $\pm\sqrt{2}$ give indefinite Hessian.

<u>Some exercises:</u>

1. $g(\underset{\sim}{x}) = \|\underset{\sim}{y} - A\underset{\sim}{x}\|^2$ where A is a p.d. invertible matrix.
2. $g(\underset{\sim}{x}) = \|\underset{\sim}{x}\|^2 + \exp\{\underset{\sim}{a}^T \underset{\sim}{x}\}$

1.3 Multivariate Normal Distribution

Multivariate normal distribution plays a key role in multivariate analysis. To model a multivariate data, mvt. normal is most commonly used. The motivations behind this predominance can be stated below.

[1] As already stated, providing a multivariate generalization to any univariate distribution may not be easy. The generalization may not be unique. For normal distribution easy extension is available.

[2] The p-mvt. normal distribution is completely specified by the first 2 moments. The interpretable parameters, and availability of efficient estimators of the parameters, make mvt. normal more popular.

[3] The fact that for jointly normally distributed variables, uncorrelatedness $\Leftrightarrow$ independence, makes verification of independence far less complicated than for normal distribution.

Defn (Multivariate Normal)

A p-variate RV $\underset{\sim}{X}$ is said to follow a mvtivariate normal distribution with parameters $\underset{\sim}{\mu}$ and $\Sigma \geq 0$ iff

$$\underset{\sim}{a}^T \underset{\sim}{X} \sim \mathcal{N}(\underset{\sim}{a}^T \underset{\sim}{\mu}, \underset{\sim}{a}^T \Sigma \underset{\sim}{a}) \text{ for all } \underset{\sim}{a} \in \mathbb{R}^p$$

Here $\underset{\sim}{\mu} \in \mathbb{R}^p$, and Σ is a $p \times p$ symmetric PSD matrix.

We write $\underset{\sim}{X} \sim \mathcal{N}(\underset{\sim}{\mu}, \Sigma)$ if $\underset{\sim}{x}$ follows a mvt. normal with parameters $\underset{\sim}{\mu}$ and Σ.

Note: $X_i \sim N(\mu_i, \sigma_{ii})$ for all $i = 1, \ldots, p$ does not imply $\underset{\sim}{X}$ is normal. Let $\underset{\sim}{e}_i$ be the vector with all zero elements except the i-th one, which is one. If $\underset{\sim}{X} \sim N_p(\underset{\sim}{\mu}, \Sigma)$ where $\underset{\sim}{\mu} = (\mu_1, \ldots, \mu_p)^\top$ and $\Sigma = ((\sigma_{ij}))$, then

$$\underset{\sim}{e}_i^\top \underset{\sim}{X} = \underset{\sim}{X}_i \sim N(\mu_i, \sigma_{ii}) \quad \forall\ i.$$

However, the following example shows a case where the marginals are normal but the joint is not. Let (X, Y) is a bivariate random variable with joint pdf

$$f(x,y) = \frac{1}{2\pi} \exp\left\{-\frac{x^2+y^2}{2}\right\} \left\{1 + xy \exp\left\{-\frac{x^2+y^2}{2}\right\}\right\}$$

Properties :

[1] Characteristic function:

Recall that the characteristic function (ch.fn.) of univariate normal with parameters η and σ^2 is $\exp(i\eta t - \frac{1}{2}t^2\sigma^2)$, at the point t. Thus,

$$\phi_{\underset{\sim}{X}}(\underset{\sim}{t}) = E\left[\exp\{i\underset{\sim}{t}^\top \underset{\sim}{X}\}\right] = E\Big[\exp\{iY\} \mid Y \sim \mathcal{N}(\underset{\sim}{t}'\underset{\sim}{\mu}, \underset{\sim}{t}'\Sigma\underset{\sim}{t})\Big] = \exp\left(i\underset{\sim}{t}'\underset{\sim}{\mu} - \frac{1}{2}\underset{\sim}{t}'\Sigma\underset{\sim}{t}\right).$$

[2] Linear transformation:

Let $\underset{\sim}{Y} = A\underset{\sim}{X} + \underset{\sim}{b}$, where A is a $q \times p$ matrix, and $\underset{\sim}{b} \in \mathbb{R}^q$. Then $\quad \underset{\sim}{y} \sim \mathcal{N}_q(A\underset{\sim}{\mu} + \underset{\sim}{b}, A\Sigma A')$

Proof: The ch fn. of $\underset{\sim}{Y}$ is

$$\begin{aligned}\phi_{\underset{\sim}{Y}}(\underset{\sim}{t}) &= E\left[\exp\{i\underset{\sim}{t}'(A\underset{\sim}{X}+\underset{\sim}{b})\}\right]\\ &= \exp\{i\underset{\sim}{t}^\top\underset{\sim}{b}\}E\left[\exp\{i\underset{\sim}{t}^\top A\underset{\sim}{X}\}\right]\\ &= \exp\{i\underset{\sim}{t}^\top\underset{\sim}{b}\}E\left[\exp\{i\underset{\sim}{s}^\top\underset{\sim}{X}\}\right] \quad \text{where } \underset{\sim}{s}=A^\top\underset{\sim}{t}\\ &= \exp\{i\underset{\sim}{t}^\top\underset{\sim}{b}\}\exp\left\{i\underset{\sim}{s}'\underset{\sim}{\mu}-\frac{1}{2}\underset{\sim}{s}'\Sigma\underset{\sim}{s}\right\}\\ &= \exp\left\{i\underset{\sim}{t}'(A\underset{\sim}{\mu}+\underset{\sim}{b})-\frac{1}{2}\underset{\sim}{t}'(A\Sigma A')\underset{\sim}{t}\right\}.\end{aligned}$$

By uniqueness property of characteristic function $\underset{\sim}{Y}\sim\mathcal{N}_q(A\underset{\sim}{\mu}+\underset{\sim}{b}, A\Sigma A^\top)$. ■

Example: $\underset{\sim}{X}\sim N(\underset{\sim}{\mu},\Sigma)$; $\underset{\sim}{Y}=A\underset{\sim}{X}+\underset{\sim}{b}$.

$$\underset{\sim}{Z}=\begin{bmatrix}\underset{\sim}{Y}\\ \underset{\sim}{X}\end{bmatrix} \qquad \underset{\sim}{Z}=C\underset{\sim}{X}+\underset{\sim}{d} \qquad C=\begin{bmatrix}A\\ I_p\end{bmatrix}_{(q+p)\times p} \qquad \underset{\sim}{d}=\begin{bmatrix}b\\ 0\end{bmatrix}$$

$$C\Sigma C^T=\begin{bmatrix}A\Sigma\\ \Sigma\end{bmatrix}\begin{bmatrix}A^T & I_p\end{bmatrix} \qquad \underset{\sim}{Z}\sim N_{q+p}\left(\begin{bmatrix}A\underset{\sim}{\mu}+\underset{\sim}{b}\\ \underset{\sim}{\mu}\end{bmatrix}, \begin{bmatrix}A\Sigma A^\top & A\Sigma\\ \Sigma A^\top & \Sigma\end{bmatrix}\right)$$

[3] Expectation and Variance. $E(\underset{\sim}{X})=\underset{\sim}{\mu}$ and $\text{Var}(\underset{\sim}{X})=\Sigma$.
Proof: By defn. $\underset{\sim}{e}_j^\top\underset{\sim}{X}=X_j\sim\mathcal{N}(\mu_j,\sigma_{jj})$ and

$$(\underset{\sim}{e}_j+\underset{\sim}{e}_k)^\top\underset{\sim}{X}=(X_j+X_k)\sim\mathcal{N}(\mu_j+\mu_k,\sigma_{jj}+\sigma_{kk}+2\sigma_{jk})$$

From the above two facts, it is clear that—

$$E(X_j)=\mu_j \quad \text{and} \quad \text{Var}(X_j)=\sigma_{jj} \quad \forall j=1,\dots,p$$

Further,

$$\text{Var}(X_j+X_k)=\text{Var}(X_j)+\text{Var}(X_k)+2\text{Cov}(X_j,X_k)=\sigma_{jj}+\sigma_{kk}+2\sigma_{jk}$$

Thus, $\text{Cov}(X_j,X_k)=\sigma_{jk}$. ■

[4] Standardization. Let $\underset{\sim}{X}\sim\mathcal{N}(\underset{\sim}{\mu},\Sigma)$, and $\Sigma=BB^T$ for some matrix B. $\Leftrightarrow \underset{\sim}{X}\overset{D}{=}B\underset{\sim}{Z}+\underset{\sim}{\mu}$, where $\underset{\sim}{Z}\sim\mathcal{N}(\underset{\sim}{0},I)$.
Proof: $[\Sigma>0]$ $\Sigma=U\Lambda U^T$ with $UU^T=U^TU=I_{p\times p}$, and $\Lambda=\text{Diag}(\lambda_1,\dots,\lambda_p)$ $\lambda_j>0\,\forall j$ as $\Sigma>0$.
Define $\Lambda^{1/2}=\text{Diag}(\sqrt{\lambda_1},\dots,\sqrt{\lambda_p})$
The choices of B can be $B=U\Lambda^{1/2}$ (non-symmetric) or $B=U\Lambda^{1/2}U^\top$ (symmetric).
In both the cases, B^{-1} exists. For symmetric B, $B^{-1}=U\Lambda^{-1/2}U^T$ where $\Lambda^{-1/2}=\text{Diag}(1/\sqrt{\lambda_1},\dots,1/\sqrt{\lambda_p})$, while for the non-symmetric B, $B^{-1}=\Lambda^{-1/2}U^T$.
($\Leftarrow$ part) $\underset{\sim}{Z}\sim N_p(0,I)$

$$B\underset{\sim}{Z}+\underset{\sim}{\mu}\sim N_p(\underset{\sim}{\mu},\ BB^T=\Sigma) \qquad \underset{\sim}{X}\overset{D}{=}B\underset{\sim}{Z}+\underset{\sim}{\mu}$$

($\Rightarrow$ part) $\underset{\sim}{X}\sim N_p(\underset{\sim}{\mu},\Sigma)$

$$\underset{\sim}{Z}=B^{-1}(\underset{\sim}{X}-\underset{\sim}{\mu})\sim N_p(0,I) \implies \underset{\sim}{X}\overset{D}{=}B\underset{\sim}{Z}+\underset{\sim}{\mu}$$

[5] ***Independence.*** Let $\underset{\sim}{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \sim \mathcal{N}_p(\underset{\sim}{\mu}, \Sigma)$, $p_1 + p_2 = p$, then $\underset{\sim}{X}_1 \perp \underset{\sim}{X}_2 \Leftrightarrow \Sigma_{12} = \underset{\sim}{0}$, where

$$\Sigma = \begin{bmatrix} \Sigma_{11}^{p_1\times p_1} & \Sigma_{12}^{p_1\times p_2} \\ \Sigma_{21}^{p_2\times p_1} & \Sigma_{22}^{p_2\times p_2} \end{bmatrix}, \quad \text{and} \quad \Sigma_{21} = \Sigma_{12}'.$$

Proof: (4 parts) We can easily verify this by property [3] of ch-fn.,
as $\phi_{\underset{\sim}{X}}(\underset{\sim}{t}) = \exp\left\{i\underset{\sim}{t}'\underset{\sim}{\mu} - \frac{1}{2}\underset{\sim}{t}'\Sigma\underset{\sim}{t}\right\} = \exp\left\{i\underset{\sim}{t}_1'\underset{\sim}{\mu} - \frac{1}{2}\underset{\sim}{t}_1'\Sigma_{11}\underset{\sim}{t}_1\right\} \times \exp\left\{i\underset{\sim}{t}_2'\underset{\sim}{\mu}_2 - \frac{1}{2}\underset{\sim}{t}_2'\Sigma_{22}\underset{\sim}{t}_2\right\}$.
As $\underset{\sim}{X}_i \sim \mathcal{N}(\underset{\sim}{\mu}_i, \Sigma_{ii})$ marginally, $i = 1, 2$, this implies

$$\phi_{\underset{\sim}{X}}(\underset{\sim}{t}) = \phi_{\underset{\sim}{X}_1}(\underset{\sim}{t}_1) \times \phi_{\underset{\sim}{X}_2}(\underset{\sim}{t}_2) \Rightarrow \underset{\sim}{X}_1 \perp \underset{\sim}{X}_2.$$

($\Rightarrow$ part) If $\underset{\sim}{X}_1 \perp \underset{\sim}{X}_2$, then $\mathrm{Cov}(X_{ij}, X_{2k}) = 0$ for each $j = 1, \ldots, p_1$, $k = 1, \ldots, p_2$. Thus, $\Sigma_{12} = 0$.

[6] ***Density.***
Case I: ($\Sigma > 0$)
By property [4], in this case $\underset{\sim}{Z} = \Sigma^{-1/2}(\underset{\sim}{X} - \underset{\sim}{\mu}) \sim \mathcal{N}(\underset{\sim}{0}, I_p)$.
By property [5], the p-components of $\underset{\sim}{Z}$ are independent and identically distributed (IID) following $\mathcal{N}(0, I)$ distribution. Thus,

$$f_{\underset{\sim}{Z}}(\underset{\sim}{z}) = (2\pi)^{-p/2} \exp\left\{-\frac{1}{2}\underset{\sim}{z}^T\underset{\sim}{z}\right\}.$$

As the transformation $\underset{\sim}{Z} \to \underset{\sim}{X} = \underset{\sim}{\mu} + \Sigma^{1/2}\underset{\sim}{Z}$ is one-one, the density of $\underset{\sim}{X}$ can be obtained as:

$$\begin{aligned} f_{\underset{\sim}{X}}(\underset{\sim}{x}) &= f_{\underset{\sim}{Z}}(\underset{\sim}{z})\Big|_{\underset{\sim}{Z}=\Sigma^{-1/2}(\underset{\sim}{X}-\underset{\sim}{\mu})} \times J_{Z\to X} \\ &= |\Sigma|^{-1/2}(2\pi)^{-p/2} \exp\left\{-\frac{1}{2}(\underset{\sim}{x} - \underset{\sim}{\mu})^T\Sigma^{-1}(\underset{\sim}{x} - \underset{\sim}{\mu})\right\}. \end{aligned}$$

as

$$J_{Z\to X} = \left|\det\left(\frac{\partial Z}{\partial X}\right)\right| = ||B^{-1}|| = \frac{1}{||B||} = \frac{1}{|B|}, \quad \text{and} \quad |\Sigma| = |BB^T| = |B|^2 \quad \Longrightarrow \frac{1}{|B|} = \frac{1}{|\Sigma|^{1/2}}.$$

Further

$$\sum_{i=1}^{p} z_i^2 = \underset{\sim}{z}^\top\underset{\sim}{z} = \left(\underset{\sim}{x} - \underset{\sim}{\mu}\right)^\top (B^{-1})^\top B^{-1}\left(\underset{\sim}{x} - \underset{\sim}{\mu}\right) = \left(\underset{\sim}{x} - \underset{\sim}{\mu}\right)^\top (BB^T)^{-1}\left(\underset{\sim}{x} - \underset{\sim}{\mu}\right) = \left(\underset{\sim}{x} - \underset{\sim}{\mu}\right)^\top \Sigma^{-1}\left(\underset{\sim}{x} - \underset{\sim}{\mu}\right).$$

Equi-probability contour plots.
The contour plots or contours are the collection of points of equal probability density. Thus, the set of points with density equal to q is:

$$S_q = \left\{\underset{\sim}{x} : f_{\underset{\sim}{x}}(\underset{\sim}{x}) = q\right\} = \left\{\underset{\sim}{x} : (\underset{\sim}{x} - \underset{\sim}{\mu})^T\Sigma^{-1}(\underset{\sim}{x} - \underset{\sim}{\mu}) = c_q\right\},$$

where

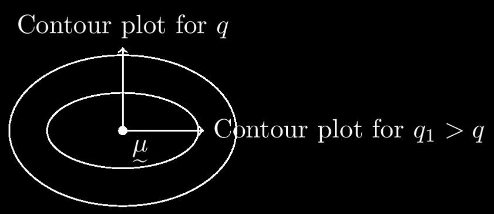
$$c_q = -2\log\left\{|\Sigma|^{1/2}(2\pi)^{p/2}q\right\}.$$

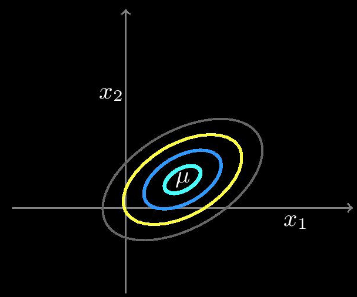
Clearly, S_q contains points on an ellipsoid, characterized by parameters $(\underset{\sim}{\mu}, \Sigma, c_q)$, centered at $\underset{\sim}{\mu}$. Also, note that the set of high density region $\rho(q) = \left\{\underset{\sim}{x} : f_{\underset{\sim}{x}}(\underset{\sim}{x}) \geq q\right\}$ is the interior of the ellipse.

For any $q_1 > q$, $S_{q_1} = \left\{\underset{\sim}{x} : (\underset{\sim}{x} - \underset{\sim}{\mu})^T \Sigma^{-1} (\underset{\sim}{x} - \underset{\sim}{\mu}) = c_{q_1}\right\}$, where $c_{q_1} < c_q$, and therefore the points form a concentric ellipsoid, with smaller lengths of principal axes.

When $\underset{\sim}{\mu} = \underset{\sim}{0}$, *and* $\Sigma = I$. In this case, $S_q = \left\{\underset{\sim}{x} : \underset{\sim}{x}^T \underset{\sim}{x} = c_q\right\}$, and thus the equi-probability contours are spherical and centered at $\underset{\sim}{0}$.

When $\underset{\sim}{\mu} \neq \underset{\sim}{0}$, *and* $\Sigma \neq I$. In this case, S_q is elliptical.

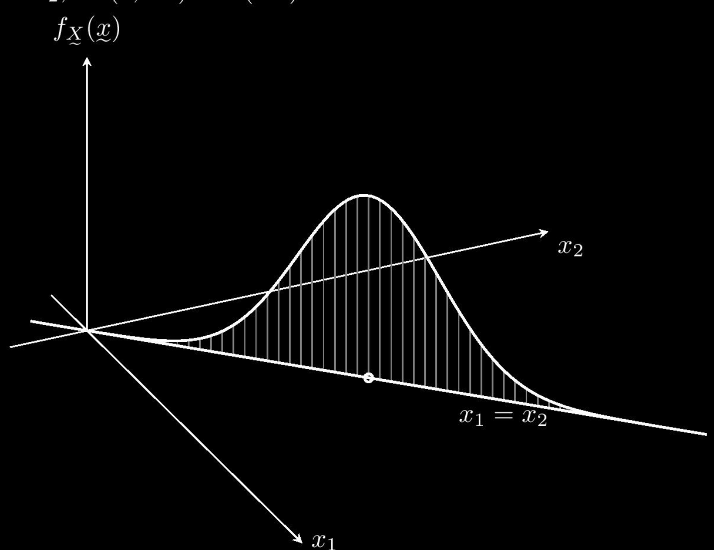




Case II: When Σ is singular, then density exists only in hyperplane which is orthogonal to the null-space of Σ, i.e., in the row-space of Σ.

An example of the case with $\det(\Sigma) = 0$**:** Let $\underset{\sim}{X} \sim \mathcal{N}\left(\begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}\right)$. Here, X_1 and X_2 are perfectly correlated. We will show that here $P(X_1 = X_2) = 1$.

Let $\underset{\sim}{a} = (1, -1)^\top$, then $\underset{\sim}{a}^\top \underset{\sim}{X} = X_1 - X_2 \sim N(0, 0)$, which implies that $E(X_1 - X_2) = 0$ and $var(X_1 - X_2) = 0$, which happen only if $P(X_1 = X_2) = 1$. Then density of $\underset{\sim}{X}$ exists on the hyperplane $x_1 = x_2$, as $(1, -1)\Sigma = (0\ 0)$.



In general, if Σ is singular, then the density will exist on the line $\Sigma y = 0$ $(y \neq 0)$, as

$$y^T X \sim N(y^T \mu, y^T \Sigma y), \quad \text{which implies} \quad P(y^T X = 0) = 1$$

.

[7] ***Independence of two linear forms.*** Let $\underset{\sim}{X} \sim N(\mu, \Sigma)$; $\quad \Sigma \geq 0$. For what choices of A and B, $\underset{\sim}{Y} = A\underset{\sim}{X} \perp\!\!\!\perp B\underset{\sim}{X} = \underset{\sim}{Z}$?

First, we see that $(\underset{\sim}{Y}, \underset{\sim}{Z})^\top$ is jointly normally distributed, as

$$\begin{bmatrix}\underset{\sim}{Y}\\ \underset{\sim}{Z}\end{bmatrix} = \begin{bmatrix}A\\ B\end{bmatrix} \underset{\sim}{X} = C\underset{\sim}{X} \sim N(C\underset{\sim}{\mu}, C\Sigma C^\top).$$

Therefore, $\underset{\sim}{Y} \perp\!\!\!\perp \underset{\sim}{Z}$, if and only if $\text{Cov}(\underset{\sim}{Y}, \underset{\sim}{Z}) = 0$.

When is $\text{Cov}(\underset{\sim}{Y}, \underset{\sim}{Z}) = 0$? Answer: $A\Sigma B' = 0$. Hence $A\Sigma B' = 0 \iff AX \perp\!\!\!\perp BX$.

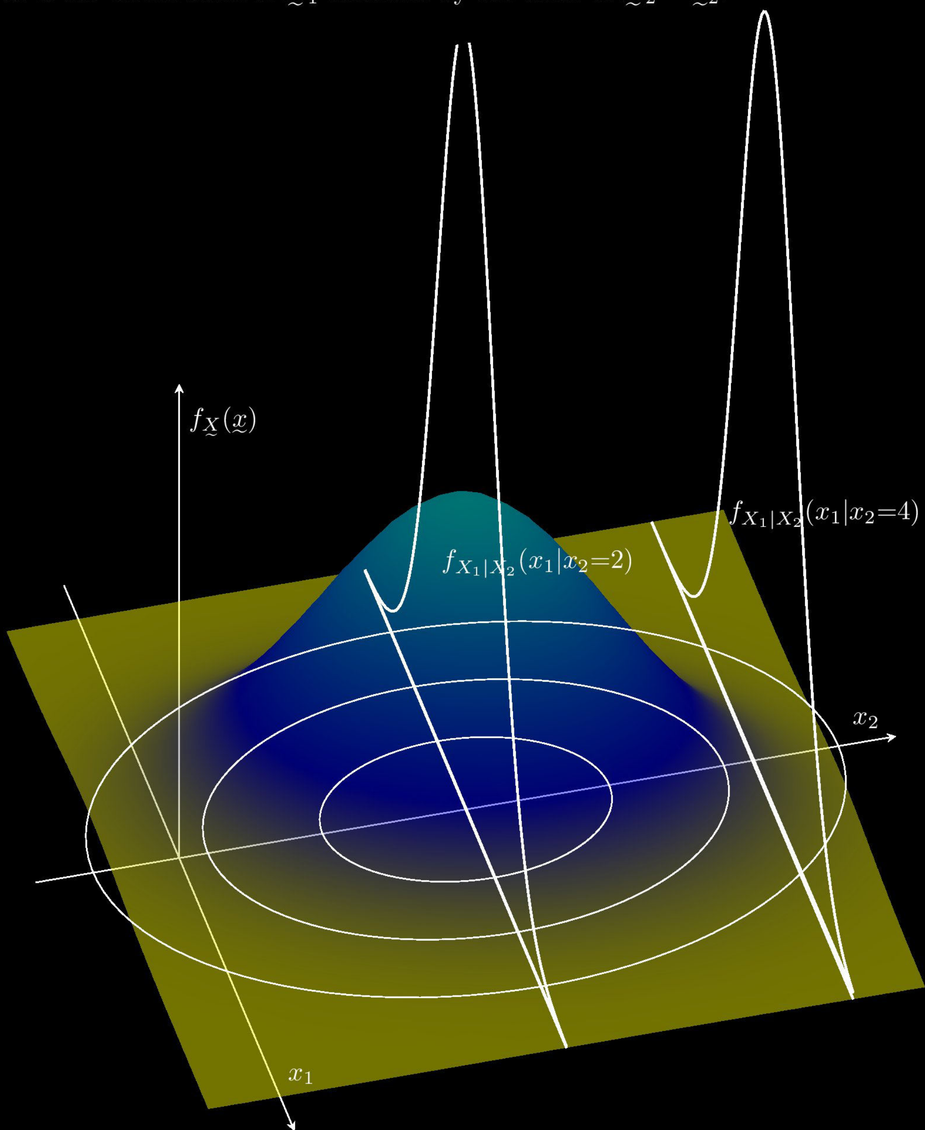
[7] (***Marginal and Conditional distributions***)
Marginal: From property (2), it is straightforward to obtain the marginal distributions. Let

$$\underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{X}_1 \\ \underset{\sim}{X}_2 \end{pmatrix},$$

then the marginal of $\underset{\sim}{X}_2 = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \underset{\sim}{X}$ is normally distributed with parameters

$$\underset{\sim}{\mu}_2 \text{ and } \Sigma_{22} = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} 0 \\ I_{p_2} \end{pmatrix}.$$

Conditional distribution: (when $\Sigma_{22} > 0$)
Question. What is the conditional distribution of $\underset{\sim}{X}_1$ given $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$?
It is the distribution of $\underset{\sim}{X}_1$ informed by the value of $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$.



To obtain this, we follow the steps below:

i. We will find B such that $\underset{\sim}{Z} = \underset{\sim}{X}_1 - B\underset{\sim}{X}_2 \perp\!\!\!\perp \underset{\sim}{X}_2$.

   As $\underset{\sim}{Z} \perp\!\!\!\perp \underset{\sim}{X}_2$, the conditional distribution of $\underset{\sim}{Z}$ given $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$ is the same as the marginal distribution.

ii. Using this fact and the marginal distribution of $\underset{\sim}{Z}$, we will find the conditional distribution of $\underset{\sim}{Z} + B\underset{\sim}{X}_2$ ($=\underset{\sim}{X}_1$) given $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$.

Step i.

$$\underset{\sim}{Z} = [I_{p_1} \;\; -B]\begin{bmatrix}\underset{\sim}{X}_1\\ \underset{\sim}{X}_2\end{bmatrix}, \qquad \underset{\sim}{X}_2 = [0 \; I_{p_2}]\begin{bmatrix}\underset{\sim}{X}_1\\ \underset{\sim}{X}_2\end{bmatrix}$$

$$[I_{p_1} \;\; -B]\begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}\begin{bmatrix}0\\ I_{p_2}\end{bmatrix} = 0 \Longleftrightarrow [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}]\begin{bmatrix}0\\ I_{p_2}\end{bmatrix} = 0 \Longleftrightarrow \quad \Sigma_{12} - B\Sigma_{22} = 0$$

$$\Longleftrightarrow \quad B = \Sigma_{12}\Sigma_{22}^{-1}$$

Therefore, for $\underset{\sim}{Z} = \underset{\sim}{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underset{\sim}{X}_2$, we have $\underset{\sim}{Z} \perp\!\!\!\perp \underset{\sim}{X}_2$, and $\underset{\sim}{Z} \sim N\big(\underset{\sim}{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underset{\sim}{\mu}_2,\; \Sigma_{11\cdot 2}\big)$, where

$$\begin{aligned}\Sigma_{11.2} &= [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}]\begin{bmatrix}I_{p_1}\\ -B'\end{bmatrix} = \Sigma_{11} - B\Sigma_{21} - \Sigma_{12}B' + B\Sigma_{22}B'\\ &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.\end{aligned}$$

Step ii. Given $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$, $\underset{\sim}{Z}$ has same distribution as its marginal. Hence, $\underset{\sim}{X}_2 = \underset{\sim}{x}_2$,

$$\underset{\sim}{X}_1 = \underset{\sim}{Z} + \Sigma_{12}\Sigma_{22}^{-1}\underset{\sim}{x}_2 \sim N\Big(\underset{\sim}{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underset{\sim}{x}_2 - \underset{\sim}{\mu}_2),\; \Sigma_{11\cdot 2}\Big)$$

1.4 Distribution of Quadratic Forms

Definition 1 (Idempotent Matrix). *A is idempotent if $A^2 = A$.*

A is symmetric idempotent $\iff$ the eigenvalues of A are 1 with multiplicity $\rho(A)$ [rank of A] and 0 with multiplicity $p - \rho(A)$.

Property: $\rho(A) = \operatorname{tr}(A)$.

Result 1. *Let $\underset{\sim}{Y} \sim N(0, I_p)$ and let A be a symmetric matrix.*

If A is idempotent then

$$\underset{\sim}{Y}^T A \underset{\sim}{Y} \sim \chi^2_{(q)} \qquad \textit{where } q = \rho(A).$$

Proof Consider the spectral decomposition of A as follows:

$$A = U \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} U^T = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} = U_1 U_1^T \quad \text{where} \quad UU^T = U^T U = I$$

$$\begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \begin{bmatrix} U_1 & U_2 \end{bmatrix} = \begin{bmatrix} U_1^T U_1 & U_1^T U_2 \\ U_2^T U_1 & U_2^T U_2 \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{p-q} \end{bmatrix} \Rightarrow \begin{cases} U_1^T U_1 = I, \\ U_1^T U_2 = 0, \quad U_2^T U_2 = I. \end{cases}$$

Now

$$\underset{\sim}{Y}^T A \underset{\sim}{Y} = (U_1^T \underset{\sim}{Y})^T (U_1^T \underset{\sim}{Y}) == \|\underset{\sim}{Z}\|^2 \qquad \text{where } \underset{\sim}{Z} = U_1^T \underset{\sim}{Y} \sim N_q(0, I)$$

Therefore,

$$= \sum_{i=1}^{q} Z_i^2 \sim \chi^2_{(q)} \qquad \iff \qquad Z_i \overset{\text{iid}}{\sim} N(0,1).$$

Example. Let $\underset{\sim}{Y} \sim N(0, \sigma^2 I)$, $\quad S_y^2 = \dfrac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ To show that $\dfrac{(n-1)S_y^2}{\sigma^2} \sim \chi^2_{(n-1)}$.

Observe that

$$(n-1)S_y^2 = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2, \qquad n\bar{Y} = \sum_{i=1}^{n} Y_i = \underset{\sim}{Y}^T \underset{\sim}{1}.$$

Therefore

$$(n-1)S_y^2 = \underset{\sim}{Y}^T \underset{\sim}{Y} - \frac{n}{n^2} \underset{\sim}{Y}^T \underset{\sim}{1}\underset{\sim}{1}^T \underset{\sim}{Y} = \underset{\sim}{Y}^T \Big(I - \frac{1}{n} \underset{\sim}{1}\underset{\sim}{1}^T \Big) \underset{\sim}{Y}.$$

Let $\left(I - \frac{1}{n} \underset{\sim}{1}\underset{\sim}{1}^T \right) = A$, then $A^2 = A$ (verify), and $\operatorname{tr}(A) = \rho(A) = q = n-1$. Therefore, by the previous theorem

$$\underset{\sim}{Z} = \frac{1}{\sigma} \underset{\sim}{Y} \sim N(\underset{\sim}{0}, I), \qquad \text{and} \qquad \frac{(n-1)S_y^2}{\sigma^2} = \underset{\sim}{Z}^T A \underset{\sim}{Z} = \frac{\underset{\sim}{Y}^T A \underset{\sim}{Y}}{\sigma^2} \sim \chi^2_{(n-1)}.$$